Jagesh
CSB046

Indian Institute of Engineering Science and Technology, Shibpur

BTech (CST) 5th Semester Examinations, November 2025

Computer Architecture and Organization -II (CS-3103)

Full marks: 50 Time: 3 hours

Answer any five

1a) Consider the following program segment

```
.
LOAD R1, MEM1
LOAD R2, MEM2
MULT R1, R2
ADD R1, R2
.
```

Prior to the first LOAD instruction, the contents of R1=1F, R2=0F, MEM1=21 and MEM2=08.

i) Show operation of the pipeline (shown in Figure 1) for the above program segment.

ii) During operand fetching for MULT (at cycle 6) what will be the content of R2?

iii) Is there any possibility of data hazard at cycle 7? 5

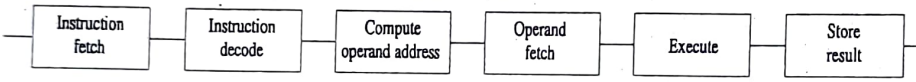


Figure 1

b) A program consists of two nested loops, with only single branch instruction at the end of each loop. The outer loop is executed 10 times and the inner loop 20 times. Determine the accuracy of following branch prediction strategies

i) Always predict taken,

ii) Always predict not taken,

iii) Use 1-bit of history (history for each loop is initialized to "taken"). 5

2a) What do you mean by `exceptions'? Explain a case of out-of-order exceptions and the methodology for handling it in the instruction pipeline. 5

b) Define the schemes 'Early restart and 'critical word first' employed in cache system design. Evaluate the performance of these two in reducing the cache miss penalty. 5

3a) Define the concept of PRAM. How an interleaved memory system approximates the structure of a PRAM? 5

b) Describe the principles of instruction execution in a superscalar-superpipelined processor with issue rate m=3 and latency = 1/2. Comment on its CPI. 5

4a) Show the network architecture of an 8-node Shuffle-Exchange SIMD m/c. If the source is 0(000) and the destination is 6(110), then state the required sequence of Shuffle/Exchange operations to circulate the data from the source to destination. 5

b) Draw a data flow graph and packaged unit of templates for Z = 5*(X+Y)*(X-Y). 5

Pg 1 of 2

5a) Consider the dual processor (Processor 1 and Processor 2) system of Figure 2 is using write-through caches. To maintain cache coherence the system uses snoopy protocol based on the write-invalidate policy. Consider the following sequence of memory access events, at different time step, where bold face numbers are for writes and the remaining are for read

Processor 1	00101120
Processor 2	01010223
Time step	12345678

i) Trace the execution of these references by the processors.
ii) Calculate the number of hits in Cache 1 and Cache 2 5

Assume both the caches are initially flushed (empty) and implement direct mapping.

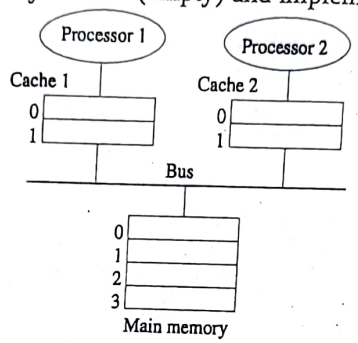


Figure 2

b) Describe directory based protocol (with full directory) to ensure cache-coherence in a multi processor system. 5

6a) State the key features of in-order issue, in-order-completion; in-order issue, out-of-order completion; and out-of-order issue, out-of-order-completion superscalar schemes. 5

b) What do you mean by loop unrolling? How does it help in ILP? Explain with an example. Consider the following loop

```
for i = 0 to n with increment 1 do
    f ();
```

and then unroll the loop by inserting 4 copies of the body f () for
i) $n = 100$,
ii) $n = 101$. 5

7. Write short notes on the following
a) Near-memory Processing, Processing-in-Memory and In-memory Computing 4
b) Non-blocking caches 3
c) Uniform and Non-Uniform Memory Access system 3

Pg 2 of 2

CSB046

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech. (CST) 5th Semester Final Examination, November 2025
Graph Algorithms (CS - 3104)

Time: 3 hours **Full Marks: 50**

Answer Q1 and any three from the rest.
Extra attempts will NOT be evaluated.
Write all parts of the same question together.
Do NOT use any specific programming language to write the pseudo-codes.

1. (a) The degree sequence of a graph is the sequence of the degrees of the vertices of the graph in nonincreasing order. Find the degree sequence of each of the following graphs: (i) C_4, (ii) $K_{2,3}$.

   (b) Use paths either to show that the graphs G and H in Fig. 1 are not isomorphic or to find an isomorphism between them.

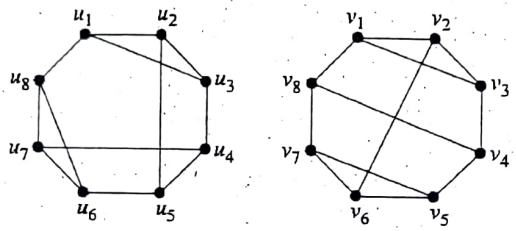


Figure 1: Test for Isomorphism

   (c) Write the chromatic numbers for the following graphs: W_3, K_4, C_5, $k_{2,3}$

   (d) Determine whether the simple graph G shown in Fig. 2(a) has an Euler circuit. Construct an Euler circuit if one exists. If no Euler circuit exists, determine whether the directed graph has an Euler path. Construct an Euler path if one exists.

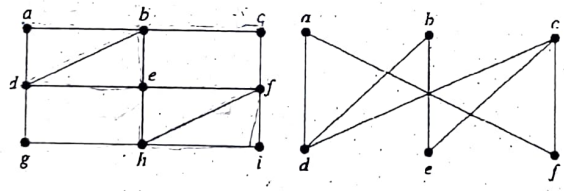


Figure 2: (a) Test for Euler Circuit/Path, (b) Test for Planarity

   (e) Write the main recurrence relation used in the *Floyd-Warshall* Algorithm. What is the running time complexity of this algorithm?

   (f) Determine whether the graph H in Fig. 2(b) is planer. If so, draw it such that no edges cross.

   (g) Fill in the blanks: The *Ford-Fulkerson* algorithm is used to find _____ and it terminates when _____.

[$7 \times 2 = 14$]

2. (a) State two differences between the Graph Coloring and the m-Colorability problems.

   (b) Write a pseudo-code for the `mColoring()` algorithm to test if the vertices of a given graph G can be colored by at most m colors. Write the complexity classes to which this algorithm belongs for the following three cases: (i) $m = 1$, (ii) $m = 2$, and (iii) $m \geq 3$.

(c) The mathematics department has six committees, each of which meets once a month. Find out the number of different meeting times that must be used to ensure that no member is scheduled to attend two meetings at the same time if the committees are C_1 = {Arlinghaus, Brand, Zaslavsky}, C_2 = {Brand, Lee, Rosen}, C_3 = {Arlinghaus, Rosen, Zaslavsky}, C_4 = {Lee, Rosen, Zaslavsky}, C_5 = {Arlinghaus, Brand}, and C_6 = {Brand, Rosen, Zaslavsky}?

$[(2 + (4 + 3) + 3 = 12]$

3. (a) State Kuratowski's theorem to check the planarity of a graph. What do we mean when we say that the two graphs G_1 and G_2 are "homeomorphic"?

(b) Write the steps involved in the elementary subdivision operation. Use a sequence of elementary subdivisions on Kuratowski's theorem to determine if the graph G in Fig. 3 is planar.

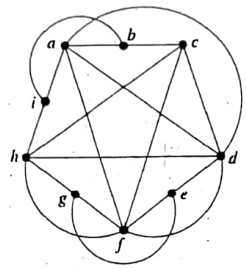


Figure 3: Kuratowski's Theorem on G

(c) Suppose that a connected planar graph has 6 vertices, each of degree 4. Into how many regions is the plane divided by a planar representation of this graph?

$[(2 + 2) + (2 + 4) + 2 = 12]$

4. (a) What happens if we try to perform topological sorting on a graph that contains a cycle? Describe the steps of the vertex deletion algorithm for topological sorting.

(b) A project has the following dependencies: Task A before B and C, Task B before D, Task C before D, and Task D before E. Draw the dependency graph and find a valid topological order using the vertex deletion algorithm.

(c) Prove the theorem: "In every DAG G, there is a node v with no incoming edges."

$[(2 + 3) + (2 + 3) + 2 = 12]$

5. (a) State and prove the "Max-flow min-cut" theorem.

(b) Mention two constraints for the maximum flow problem. Write a pseudocode for the *Ford-Fulkerson* algorithm for finding the maximum flow f in a flow network G. What is the running time complexity of this algorithm?

(c) Consider the following flow network G given in the following table. Use the Ford-Fulkerson algorithm to find the maximum flow from s to t. Show all augmenting paths, flow values, and residual capacities after each iteration.

Edge	Capacity
$s \to a$	10
$s \to b$	5
$a \to b$	15
$a \to t$	10
$b \to t$	10

$[3 + (2 + 2 + 1) + 4 = 12]$

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech (CS) 5th Semester End-Semester Examinations, Nov 2025

Jayash
CSB046.

Database Management Systems (CS 3102)

Full Marks: 50 **Time: 3 hrs**

Answer Question No. 1 and any two from the rest.

1. Define the following:

a) Self relationship
b) B-Tree
c) Full functional dependency
d) Serializability
e) Lock compatibility matrix

[5x2]

2. a) Define domain dependency and data dependency. Give example for each of these types of dependencies.

b) Consider the following set of functional dependencies
{AB → C D → EG C → A BE → C
BC → D CG → BD ACD → B CE → AG}
Prove that BD is a candidate key.

c) Consider a relation RENT_DUE (qtr#, emp_id, date_of_occupancy, rent) with fds as follows :
qtr# date_of_occupancy → emp_id rent
emp_id → qtr#

Is it in 2NF, 3NF or BCNF ? Justify your answer.

[5+5+10]

3. A library database consisting of the following relations :

BOOK(title, author, p_name, lc_no)
BORROWER(name, addr, city, card_no)
LOAN(card_no, lc_no, date)

a) Write a query in relational algebra to list the books (title) that was borrowed before 1/1/2024.
b) Draw the parse tree of the above query.
c) Optimize the query step by step and modify the parse tree accordingly.

[5+5+10]

4. In the context of transaction management, answer the following:

a) What is atomicity property? Which module takes care to maintain this property?

b) What is lost-update problem? Which property of transaction management is mostly violated in such a problem?

c) Define Conflicting operations.

d) What is Wait-Die scheme? Is it a deadlock avoidance or deadlock prevention scheme? Justify your answer.

[4x5]

5. Write short notes on the following:

a) Lossless Join Decomposition into 3NF
b) Log based recovery techniques

[10+10]

2023CSB046
Rupesh Jaiswal.

Indian Institute of Engineering Science and Technology, Shibpur

B. Tech. (CST) 5th Semester End-Term Examination, November, 2025

Microprocessor Based Systems (CS 3101)

Full Marks: 50 Time: 3 Hours

- **Attempt any five (5) questions.**
- **Answers should be precise, to the point, and in your own words as far as practicable.**
- **If not explicitly mentioned, assume Intel 8085A as the base microprocessor.**
- **Make your own assumptions, if necessary, and state them at proper places.**

1. (a) Write an 8085 assembly language subroutine "**void mem_swap(void *p1, void *p2, int n)**" that swaps the contents of **n** bytes starting at the address **p1** with that starting at the address **p2**.

   (b) Let a typical 8085 SDK has to be interfaced with a button so that, whenever the button is pressed, memory locations starting from the address 2000H upto 2FFFH are intialized with 00H. Explain with suitable diagram(s) how you can achieve this.

   [5+5]

2. (a) Explain the roles of Assembler Directives, BIOS Calls, DOS Calls in assembly language programs targetted for MASM Assembler running on MSDOS 6.22 at 8086 Based IBM PC Microcomputers.

   (b) Write an 8086 assembly language subroutine "**void sort(int a[ ], int n)**" that sorts first **n** elements of **a**[ ], an array of 16-bit integers, in ascending order.

   [5+5]

3. (a) What is **Based Indexed Addressing Mode** and how is it useful? Show how a typical 8086 instruction, which uses **Based Indexed Addressing Mode**, can be realized using 8085 instruction(s).

   (b) Explain with suitable diagram(s) the advantages of adopting **Harvard Architecture** in Microcontrollers (Eg, 8051), and SOCs (Eg, NodeMCU).

   [5+5]

4. Say, there is an 8086 based IBM PC compatible micro-computer running MSDOS 6.22 Operating System. Explain with suitable diagram(s) the working principle(s) of how some additional IO device (say, NIC or Sound Card) can be installed (interfaced) along with its "driver" (provided by the manufacturer of the IO device) so that application programs can be writen to "operate" on the IO device.

   [10]

5. (a) Explain with suitable timing diagram how the READY signal (pin) of 8085 processor can be used to interface and synchonze with slower memory devices (chips).

   (b) Explain with suitable example(s) how the hardware interrupt line **INTR** works in 8085 microprocessor and how it can be used to interface IO Devices.

   [5+5]

6. Write short notes on the following.

   (a) **Segmented Memory Model** of 8086 Microprocessor.

   (b) **Arduino IDE** for **NodeMCU (ESP8266)**.

   [5+5]

Indian Institute of Engineering Science and Technology, Shibpur

B. Tech (CST) 5th Semester End-Semester Examination, November, 2025

Web Technology (CS3122)

Tapash
CSB046

Full Marks: 50 *Time: 3 Hours*

- *Answer all questions.*
- *Answer to all parts of a same question together.*
- *In case of typing mistake/error/confusion in the question paper: Don't panic. Don't call invigilators out. You will get the deserved marks based on an honest attempt.*
- *Provide diagram wherever necessary*



1. Choose the correct alternative. No explanation is needed for your answer

(i) Which of the following best describes "slicing design" in web development?

a) Translating a webpage into smaller, responsive sections only for portable devices.

b) Optimizing image files for faster loading times on a webpage.

c) Breaking down a complete design into several individual HTML/CSS components and image assets.

d) Creating interactive animations for specific elements on a webpage.

(ii) What is the primary purpose of using "stock media" in web design?

a) To create completely unique and custom visual content for every website.

b) To save time and money with pre-existing and licenced photos, videos, audio, and graphics for web development.

c) To ensure all website contents are authentic.

d) To generate random images and videos for a dynamic user experience.

(iii) A "navigation website storyboard" primarily focuses on:

a) Designing the visual aesthetics of a website's menu bar.

b) Mapping out the user's journey and movement between different website pages.

c) Planning the content strategy for a website's blog section.

d) Creating a timeline for the entire website development process.

(iv) How do "cloud services" benefit web technology?

a) They provide physical servers and hardware for website hosting.

b) They eliminate the need for any internet connection to access websites.

c) They offer on-demand computing resources, enabling flexible and scalable hosting without managing physical infrastructure.

d) They automate the creation of all website content.

(v) Which of the following is considered part of the "attack surface of a webpage"?

a) The website's favicon.

b) User input fields (e.g., login forms, search bars).

c) The background image of the webpage.

d) The font used for the webpage's text.

[5]

2. Read the paragraph and Clearly write True or False for the following statements

Advanced JavaScript investigates beyond the basics to empower developers with robust and efficient code. Key topics include closures, where a function remembers and accesses variables from its outer scope even after the outer function has finished execution. Asynchronous programming is crucial for non-blocking operations, utilizing Promises and the more modern async/await syntax (e.g., fetch API). Understanding the Event Loop is vital for comprehending how JavaScript handles concurrent tasks. Other advanced concepts involve prototypal inheritance (the core of JavaScript's object model), higher-order functions like map (), filter (), and reduce (), which operate on or return other functions, and functional programming paradigms for cleaner, more predictable code. Unlike classical inheritance that uses classes to derive other classes, prototypal inheritance relies on objects directly inheriting from other objects known as prototypes. Mastering these topics is essential for building complex, scalable web applications.

(i) Closures allow a function to forget variables from its outer scope once the outer function has finished executing. F

(ii) Syntaxes like 'async' and 'await' are primarily used for synchronous programming in JavaScript. F

(iii) Understanding the Event Loop is important for comprehending how JavaScript handles concurrent tasks. T

(iv) JavaScript's object model is based on classical inheritance, similar to languages like Java. F

(v) The functions like map (), filter (), and reduce () are examples of higher-order functions in JavaScript. T

[5]

3.Read the paragraph and match the following. No marks will be awarded for partially correct matching.

Secure website development is paramount in today's digital landscape. It involves a multi-faceted approach, starting with secure coding practices to prevent common vulnerabilities like SQL injection and Cross-Site Scripting (XSS) through rigorous input validation and output encoding. Implementing HTTPS and TLS encryption is fundamental to protect data in transit, ensuring secure communication between client and server. Robust authentication and authorization mechanisms, including multi-factor authentication, restrict access to sensitive resources. Regular security audits, penetration testing, and keeping all software and libraries up-to-date are crucial for proactively identifying and addressing potential weaknesses, minimizing the attack surface.

1. Input Validation & Output Encoding	a. Limiting access to sensitive resources (3)
2. HTTPS and TLS Encryption	b. Proactively identifies and addresses weaknesses (4)
3. Multi-Factor Authentication	c. Prevents common vulnerabilities like SQL Injection (1)
4. Security Audits & Penetration Testing	d. Protects data in transit (2)
5. Keeping Software Up-to-Date	e. Minimizes the attack surface (5)

[5]

4. Read the paragraph and answer the following questions in one sentence only.

JAMstack (JavaScript, APIs, and Markup) represents a modern web development architecture designed for speed, security, and scalability. Instead of traditional monolithic setups where content is dynamically generated by a server on each request, JAMstack websites are primarily pre-built into static HTML, CSS, and JavaScript files. These assets are then served directly from a Content Delivery Network (CDN), significantly reducing load times and improving performance globally. Dynamic functionalities, like forms or user authentication, are handled by JavaScript interacting with various third-party APIs or serverless functions, effectively decoupling the frontend from complex backend logic. This approach offers enhanced security by minimizing the server-side attack surface and simplifies deployments

(i) In JAMstack, what is the full form of JAM?

(ii) "JAMstack websites primarily focus on back end languages"-Do you agree?

(iii)Where are JAMstack website assets primarily served from?

(iv)What is a security benefit of the JAMstack approach mentioned in the paragraph?

(v) How are dynamic functionalities handled in JAMstack?

[5]

5. Explain the meaning of following terms in not more than 2-3 sentences

(i) Network Protocols (ii) Client-server architecture (iii) Structuring, styling, and interactive functionalities on web page (iv) HTTP and HTTPS (v) Frameworks in front end development (vi) Caching for improving user experience (vii) Responsive Web Design (RWD) (viii) Progressive web apps (PWAs) (ix) Serverless architecture (x) Web Assembly

[10]

6 Write short note on

(i) Spoofing attack (ii) Whitelist validation

[5+5=10]

7. Answer the following questions

(i) Write three benefits of AI based product recommender.

(ii) Explain tree diagram representing site structure

(iii) Mention four features of NodeJS

[3+3+4=10]